I am Joe's Heart

I'm certainly no Beauty. I weigh 400grams, am red-brown colour and have an unimpressive shape. I am the dedicated slave of you. I hang by ligaments in the centre of his chest. I am about 15 cm long and, at my widest point, 10 cm across-more pear-shaped than valentine. Whatever you may have heard about me from poets, I am really not a very romantic character. I am just a hard working four-chambered pump-actually two pumps, one to move blood to the lungs, the other to push it out and into the body. Every day, I would say, I pump blood through 96,000 kilometres of blood vessels. That's enough pumping to fill an 18,000 litre tank car.

When Joe thinks of me at all, he thinks of me as fragile and delicate. Delicate! When so far in his life I have pumped more than 300,000 tons of blood? I work twice as hard as the leg muscles of a dash runner, or the arm muscle of a heavyweight boxing champ. Let them try to go at my pace and they would turn to jelly in minutes. No muscles in the body are as strong as I am-except those of a woman's uterus as she expels a baby. But uterine muscles don't keep at in day and night for 70 years, as I am expected to do.

That, of course, is a slight exaggeration. I do rest-between beats. It takes about three tenths of a second for my big left ventricle to contract and push blood out into the body. Then I have a rest period of half a second.

Joe almost never thinks of me-which is good. I don't want him to become one of those heart neurotics and worry us both into real trouble. When he does worry about me it is almost always about the wrongs things. One night, as he was drifting off to sleep, Joe was listening to my quiet thumping-that's the opening and closing of my valves-and he thought he heard me "skip" a beat. He was quite worried. Was I giving out on him? He needn't have been concerned.

From time to time, my ignition system gets momentarily out of whack-just like the ignition system on Joe's car. I generate my own electricity , and send out impulses to trigger contraction. But occasionally I will misfire, piling one beat on top of another. It sounds as if I have "skipped"-but I haven't. Joe would be surprised how often this happens when he isn't listening.

After a nightmare he sometimes wakes up and worries because I am racing. That's because when he runs for his life in his dreams, I run too. Joe's worries actually aggravates things-makes me go still faster. If he would calm down, so would I. But if he can't, there is a way to slow me down. The vagus nerves act as a brake. They pass up through the neck-behind the ears, at the hinge of the jaws. Gentle massage here will slow my beat.

Joe blames almost everything on me-fatigue, dizzy spells and such. But I have little to do with his fatigue, and his occasional dizzy spells usually trace to his ears. From time to time he will be doing his desk work, and will get a sharp pain in the

chest. He fears that he is about to have a heart attack. He needn't worry. That pain comes from his digestive tract-payment for the heavy meal eaten a couple of hours earlier. When I am in trouble, I usually send out a pain signal only after undue exertion of emotion. That's the way I tell him I am not getting enough nourishment to carry the work he is loading on me.

Joe had a heart attack five years ago and didn't even know it. He was too busy to notice that tiny twinge of pain in his chest. The artery that clogged was a small one on my rear wall. It took me two weeks to sweep away the dead tissue and repave the area with a scar not much larger than a pea.

Joe comes from a family where heart disease has occurred often, so statistics say that I am going to give him trouble, too. Of course, he can't do anything about his heredity. But he can do a lot to minimize the risk.

Let's start with overweight.
Joe is getting extra padding around the middle and jokes about his middle-age spread. It's no laughing matter. Every kilogram of his excess fat contains something like 640 km of capillaries which I have to push blood through. And that is in addition to the work of carrying around each extra pound.

And that brings me to Joe's blood pressure. It's 140/90-the upper limit of normal for his age. The 140 measures the pressure I work against while contracting, and the 90 is the pressure while I am resting between beats. the lower figure is the more important. The higher that figure rises, the less rest I get. And without adequate rest a heart simply works itself to death.

Joe could give me a break in other ways. He is a competitive, driving, worrying sort of fellow-you know, the successful-businessman type. He doesn't realize that his constant fretting continually stimulates his adrenal glands to produce more adrenaline and noradrenalin. If Joe relaxes, I relax.

Exercise is another thing. Joe is one of those weekend athletes who take it in big doses. He still like that rushing-up-to-the-net bit in tennis, as if he were a college boy. When he does this, my work load goes up as high as five times normal. What Joe should be doing is taking regular, mild exercise. A walk of a mile or two a day would help. Climbing a couple of flights of stairs to his office wouldn't hurt either. His office is on the tenth floor, but he could walk up the first two flights and then catch the elevator. Little things like that would do a lot. Fatty deposits are already beginning to block some of my arteries. This regular exercise causes new blood pathways to develop. Then if one artery closes down there are others to nourish me.

I'm not the demanding type. I'll do the best I can for Joe under any circumstances. Just the same, there are those breaks he could give me: slim down a bit, take regular exercises, relax a little more, cut down on fats and smoking. If he would only do those things, I could keep on working for Joe for a long time.

Questions

Using the above article, your text and general knowledge answer the following questions.

1. Why do animals need a heart?

2. It has often been said your heart is the size of your fist. Use the information in the above and a ruler to find if this is accurate. Write a comment to explain what you found.

3. Why is your pulse rate not constant?

4. Why do you think fit athletes have a lower resting heart rate?

5. Some say we love with our hearts. Explain why you think this is an unreasonable suggestion.

6. What is a heart attack?

7. Why do you think a person's blood pressure increases as they gain weigh?

8. Why does worrying negatively affect your heart? (use the term adrenalin in your answer)

9. What causes a heart to beat?

10. Draw the flow of blood through the heart on the diagram below and label as many of the parts of the heart you can.

